Chapter V

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Differentiation and Power Series

1 Introductions and Motivation

Similar to the definitions of continuity and limit, the formal definition of a derivative came long after it was being utilised in calculations. In fact, the notion of a derivative was formalised long before that of continuity, despite differentiability being a sufficient condition for continuity. This work was primarily done by Cauchy in an attempt to prove a particular theorem that is now known as the *mean value theorem.* This theorem says that the average rate of change of the function over a closed interval is equal to the instantaneous rate of change at some point in the open interval. Cauchy was the first person to realise the importance of this theorem and its wrestle with its proof. In particular, Cauchy realised that this theorem was merely a stepping stone to the proof of a more general theorem that allowed one to relate functions to *power series.* That is, series of the form

$$\sum_{k=0}^{\infty} a_k x^k.$$

It was the belief for some time that all functions could be expressed as power series and it was Cauchy's realisation that this claim was intimately related to the mean value theorem that led to the foundations of analysis. You may wonder why you would even want to write a function as a power series in the first place. At first glance, surely analysing infinite series of polynomials is much more complicated than analysing the function directly. In fact, this is to the contrary, since it appeared to be the case that power series behaved incredibly nicely with the operations of calculus. To demonstrate this point, consider the follow power series

$$1 + x + x^2 + x^3 + \dots.$$

This is the well known geometric series introduced in chapter 3, and was shown to converge providing $x \in (-1, 1)$. In that case, the series was shown to have sum

$$\frac{1}{1+x} = 1 + x + x^2 + x^3 + \dots.$$

Substituting in $-x^2$ for x in this expression gives the series

$$\frac{1}{1-x^2} = 1 - x^2 + x^4 - x^6 + \dots.$$

It is well known that the derivative of $\arctan(x)$ is $1/(1-x^2)$ and $\arctan(0) = 0$. Assuming that the integral of a power series works the same as the integral over a finite summation, that is that integrates term by term, we can arrive at the expression

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots.$$

Now setting $x = 1$, and using the fact that $\arctan(1) = \pi/4$, we find that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots.$$

It is emphasised here that none of these arguments are rigorous, and certainly do not constitute a proof. Along the way, huge jumps and assumptions have been made in how operations like integration behave with infinite series of polynomials. Yet, this result that has arisen can at least be verified upto come degree of accuracy. That is, by computing the partial sums of the derived series directly, they can be shown to be getting closer and closer to $\pi/4$. This is remarkable. Despite these arguments being on shaky grounds, practical and insightful results can be derived. This assumption that power series were well behaved with the operations of calculus continued to provide results to long standing unsolved mathematics problems, and further, they appeared to be correct. One such example is the solution to the famous Basel problem, namely the value of the sum of the series

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

This was famously solved by Euler using similar methods to those discussed above. He managed to show, albeit in a non-rigourous manner as according to today's standards, that the value of the sum is given by

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

So whilst it was purely an assumption of how derivatives and integrals interacted with power series, the fact that this assumption kept leading to apparently correct results seemed to heavily suggest that it was indeed true. This was the belief for a good 200 years. Yet, the discovery of the Fourier series

$$1 = \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{2k-1} \cos\left(\frac{(2k-1)\pi x}{2}\right)$$

seemed to put a spanner in the works of this understanding. In particular, if this series were true, it would seem reasonable that it could also be differentiated term by term in a similar manner to how power series were assumed to be differentiated. Doing this, we find that

$$0 = 2 \sum_{k=1}^{\infty} (-1)^k \sin\left(\frac{(2k-1)\pi x}{2}\right).$$

This series now only converges for when x is an even integer. So how can it be that derivatives are so well behaved with power series, but not for series of sine and cosine? What is so special about power series? This is one of the questions we hope to begin to explore in this chapter.

In the end though, the interaction between power series and calculus is not going to be entirely useful given that the only power series we know converges is the geometric series. Being able to differentiate and integrate the power series for the

geometric series can only get us so far. However, results such as the power series for $\arctan(x)$ that was 'derived' above, perhaps lead us to the suspicion that all functions could be written as power series. In fact, it was claimed by Euler and Lagrange that this is indeed the case and could be done via so called *Taylor series*. Taylor gave an explicit formulae for writing a given function as a power series, or at least is appeared so. If this is indeed the case, this would give us an extremely powerful tool for differentiating, as well as integrating, practically every function. In turn, given that dynamical systems are modelled by differential equations, and solving those equations often amounts to integrating potentially complicated functions, this tool would allow us to solve practical real world problems.

To conclude, in this chapter we are concerned with the interplay between power series and general functions. We have seen that this interplay yielded insightful and practical results. This interplay is then separated into two parts. Firstly, we have seen that by assuming power series behave well with respect to differentiation, and integration, this led to insightful results. Yet, the same cannot be said when derivatives are applied to different series of functions, namely Fourier series. Why is it then that power series are so robust to derivatives and integrals? We will not address the integral point in this course, since integration is left to the second semester, but we will address the derivative component. The second, and most crucial point we wish to address, is whether all functions can be written as power series, and if so, how can we construct the power series from a given function? If this process exists, it certainly will be intimately related to this first point of how derivatives interact with power series, since the coefficients of the corresponding power series would be related to the derivative, or perhaps higher order derivatives, of the function at a specified point.

2 Differentiable Functions

Given the opening discussion, both the derivative of power series and writing functions as power series, evidently require a notion of a derivative of a function. Without a rigorous footing of what this means, we don't stand a chance to analyse the questions laid out in the opening section. With this in mind, the rigorous definition of the derivative of a function at a point it given below.

Definition 2.1: Differentiable Functions

Let $f : D \to \mathbb{R}$. Then f is said to be ***differentiable at the point*** $c \in D$ if the limit

$$f'(c) := \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists. The number $f'(c)$ is called the ***derivative of*** f ***at*** c**.**

Whilst this definition is perfectly valid, it is often recast into an equivalent form that is useful for calculations. This is outlined in the following remark.

Remark 2.2: Differentiable Functions

The limit in Definition 2.1 can also be rewritten as

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}.$$

This is done by simply shifting the function according the the transformation $h = x - c$. It turns out that this form of the limit can be useful for simplifying calculations.

Utilising this remark, we are now able to prove the derivative of some well known functions, namely polynomials.

Example 2.3: Differentiable Functions

(i) The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 1$ is differentiable at each $c \in \mathbb{R}$.

Proof. Let $c \in \mathbb{R}$. Then

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = \lim_{h \to 0} \frac{1 - 1}{h} = 0.$$

Hence, $f'(c)$ exists for each $c \in \mathbb{R}$ and further $f'(c) = 0$. □

(ii) The function $g : \mathbb{R} \to \mathbb{R}$ given by $g(x) = x$ is differentiable at each $c \in \mathbb{R}$.

Proof. Let $c \in \mathbb{R}$. Then

$$\lim_{h \to 0} \frac{g(c+h) - g(c)}{h} = \lim_{h \to 0} \frac{c + h - c}{h} = 1.$$

Hence, $g'(c)$ exists for each $c \in \mathbb{R}$ and further $g'(c) = 1$. □

(iii) The function $r : \mathbb{R} \to \mathbb{R}$ given by $r(x) = x^2$ is differentiable at each $c \in \mathbb{R}$.

Proof. Let $c \in \mathbb{R}$. Then

$$\lim_{h \to 0} \frac{r(c+h) - r(c)}{h} = \lim_{h \to 0} \frac{(c+h)^2 - c^2}{h} = 2c.$$

Hence, $r'(c)$ exists for each $c \in \mathbb{R}$ and further $r'(c) = 2c$. □

Similar to continuous functions, functions are rarely differentiable at a single point. Instead, they are differentiable on some interval which can be a subset of the functions domain. This definition is outlined below.

Definition 2.4: Differentiable on Interval

Let $f : D \to \mathbb{R}$. Then f is said to be ***differentiable on the interval*** $I \subset D$ if f is differentiable at each point $c \in I$.

3 Properties of Differentiable Functions

Mirroring the previous chapters, when introducing a new concept or definition, in this case the derivative of a function, we wish to see if this property is preserved under the operations on $\mathbb{R}$ as well as further properties emerging from its differentiability. Given that the definition of a derivative involves the notion of limit, it would seem sensible that it is related to continuity. This is outlined in the following theorem.

Theorem 3.1: Differentiable implies Continuous

If $f : D \to \mathbb{R}$ is differentiable at the point $c \in D$, then it is continuous at c.

Proof. Let f be differentiable at $c \in D$. Then,

$$\lim_{x \to c} \left(\frac{f(x) - f(c)}{x - c} \right) = f'(c)$$

Hence,

$$\begin{aligned} \lim_{x \to c} (f(x) - f(c)) &= \lim_{x \to c} \left(\frac{f(x) - f(c)}{x - c} \right) \lim_{x \to c} (x - c) \\ &= f'(c) \cdot 0 \\ &= 0 \end{aligned}$$

Hence, $\lim_{x \to c} f(x) = f(c)$ and thus $f(x)$ is continuous at the point $c \in D$. □

This theorem tells us that the derivative of a function at a point is sufficient for proving continuity at a point. An obvious question to ask is if the converse is true. That is, must all continuous functions be differentiable? This is not true, and is demonstrated by the following canonical counterexample.

Example 3.2: Absolute Value Function

Let $|\cdot| : \mathbb{R} \to \mathbb{R}$ be the absolute value function on $\mathbb{R}$. Then $|\cdot|$ is continuous on $\mathbb{R}$ but not differentiable at 0.

Proof. The proof that $|\cdot|$ is continuous on $\mathbb{R}$ is left as an exercise. Consider now the right sided limit of the derivative at 0. This is given by

$$\lim_{h \to 0^+} \left(\frac{|h| - |0|}{h} \right) = \lim_{h \to 0^+} \left(\frac{h}{h} \right) = 1.$$

Similarly, the left sided limit of the derivative at 0 is given by

$$\lim_{h \to 0^-} \left(\frac{|h| - |0|}{h} \right) = \lim_{h \to 0^-} \left(\frac{-h}{h} \right) = -1.$$

Since the two limits are not equal it follows that

$$\lim_{h \to 0} \left(\frac{|h| - |0|}{h} \right)$$

does not exist. Thus, $|\cdot|$ is not differentiable at 0. □

So clearly not all continuous functions are differentiable. If one prefers to think geometrically, a function can be seen to be not differentiable at a point if that point corresponds to a cusp. This is precisely the case for the absolute value function. All that remains to show now is how differentiable functions interact with the algebraic operations on $\mathbb{R}$.

Theorem 3.3: Algebraic Properties of Differentiable Functions

Let $f : D \to \mathbb{R}$ and $g : D \to \mathbb{R}$ both be differentiable at the point $c \in D$. Then,

(i) $(f+g)'(c) = f'(c) + g'(c)$.

(ii) $(\lambda f)'(c) = \lambda f'(c)$ for each $\lambda \in \mathbb{R}$.

Proof. (i) Since f and g are differentiable at $c \in D$ it follows by the algebraic theorem of limits that

$$\begin{aligned}\lim_{x\to c}\left(\frac{f(x)+g(x)-f(c)-g(c)}{x-c}\right) &= \lim_{x\to c}\left(\frac{f(x)-f(c)}{x-c}\right) + \lim_{x\to c}\left(\frac{g(x)-g(c)}{x-c}\right)\\ &= f'(c)+g'(c).\end{aligned}$$

□

Proof. (ii) *Exercise* □

This theorem is hopefully well known and has been utilised in many of your previous calculus courses. We will not outline examples of this theorem being utilised here since it is not the purpose of this chapter to produce derivatives of specific functions. Instead, it is the proof of this theorem that is emphasised based on the new rigorous definition of a derivative. Indeed, we expect such a theorem to be true, and we can show it is consistent with this definition of a derivative.
It is also worth mentioning that this theorem, like for continuous functions, is related to the topic of linear algebra since it shows that the space of differentiable functions on $\mathbb{R}$ forms a subspace of the space of continuous functions on $\mathbb{R}$.
The next theorem to verify is the product rule of differentiable functions. This is outlined and proven below.

Theorem 3.4: Product Rule

Let $f : D \to \mathbb{R}$ and $g : D \to \mathbb{R}$ both be differentiable at the point $c \in D$. Then,

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

Proof. A short calculation shows that

$$\frac{f(x)g(x) - f(c)g(c)}{x - c} = \frac{f(x)g(x) + f(x)g(c) - f(x)g(c) - f(c)g(c)}{x - c}.$$

Utilising this expression, the algebra of limits and Theorem 3.1, it follows that

$$\begin{aligned}\lim_{x\to c}\left(\frac{f(x)g(x) - f(c)g(c)}{x - c}\right) &= \lim_{x\to c}(f(x)) \lim_{x\to c}\left(\frac{g(x) - g(c)}{x - c}\right) \\ &\quad + g(c) \lim_{x\to c}\left(\frac{f(x) - f(c)}{x - c}\right) \\ &= f(c)g'(c) + g(c)f'(c).\end{aligned}$$

□

Again, this theorem is well known and has been utilised in several previous courses. Given that the primary aim of this chapter is to relate differentiation to power series, it would be fruitful to first produce the derivative of a general polynomial of degree n. Fortunately, this can be done by simply utilising the product rule and Theorem 3.3. This is outlined in the following example.

Example 3.5: Polynomials

Let $p : \mathbb{R} \to \mathbb{R}$ given by

$$p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$$

where $a_i \in \mathbb{R}$ for each $i \in \{1, \ldots, n\}$. From Example 2.3, and utilising Theorem 3.3 and Theorem 3.4, it follows that p is differentiable at each point $c \in \mathbb{R}$ with derivative given by

$$p'(x) = na_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + a_1.$$

A small extension of the product rule address the derivative of the division of two differentiable functions. This is formally known as the *quotient rule*. The theorem is outlined below and the proof is left as an exercise to the reader.

Theorem 3.6: Quotient Rule

Let $f : D \to \mathbb{R}$ and $g : D \to \mathbb{R}$ both be differentiable at the point $c \in D$ and $g(x) \neq 0$ for each $x \in D$. Then,

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{[g(c)]^2}.$$

Proof. Exercise □

The final property of differentiable functions worth analysing is their behaviour under function composition. This is known as the *chain rule* and essentially utilises the fact that differentiable functions are continuous as well as the fact that the com-

position of continuous functions are continuous. The statement of the theorem and proof is outlined above.

Theorem 3.7: Chain Rule

Let $f : D_1 \to \mathbb{R}$ and $g : D_2 \to \mathbb{R}$ with $f(D_1) \subset D_2$. Then, if f is differentiable at $c \in D_1$ and g is differentiable at $f(c) \in D_2$, the function $h : D_1 \to \mathbb{R}$ given by $h(x) = g(f(x))$ is differentiable at c with

$$h'(c) = f'(c)g'(f(c)).$$

Proof. Define the function $d : D_2 \to \mathbb{R}$ by

$$d(y) = \begin{cases} \dfrac{g(y) - g(f(c))}{y - f(c)} & : y \neq f(c) \\ g'(f(c)) & : y = f(c) \end{cases}.$$

It follows that for each $x \in D_1$ we have

$$\frac{g(f(x)) - g(f(c))}{x - c} = d(f(x))\frac{f(x) - f(c)}{x - c}.$$

Hence, by the algebra of limits, continuity of composite functions and Theorem 3.1, we have that

$$\begin{aligned} h'(c) &= \lim_{x \to c} \left(d(f(x)) \frac{f(x) - f(c)}{x - c} \right) \\ &= \lim_{x \to c} (d(f(x)) \lim_{x \to c} \left(\frac{f(x) - f(c)}{x - c} \right) \\ &= f'(c)g'(f(c)). \end{aligned}$$

□

This concludes all the proofs of the standard properties of differentiable functions. As previously mentioned, many of these theorems will have been utilised to full effect in previous courses to produce interesting and insightful results. The purpose of this section demonstrates that indeed each of these theorems are valid according to the rigorous definition of a derivative.

4 Interior and Mean Value Theorem

Recall that the primary aim of this section is to relate so called power series to differentiable functions. It was Cauchy that realised that should a relationship exist, it must be intimately to a theorem known as the *mean value theorem*, which we will prove in this section. In order to prove this though, we are first required to prove some earlier other theorems to arrive at our desired result. The first of these theorems is the so called interior value theorem that was discovered by Fermat way before the formal work of real analysis was developed, and in fact even before the work of Newton and Leibniz. It certainly wasn't written in the form we will present it, but the essence of the theorem was nonetheless discovered by Fermat. For this

reason, the theorem is sometimes also known as *Fermat's theorem.* The statement and proof of the theorem is given as follows.

Theorem 4.1: Interior Extremum Theorem/Fermat's Theorem

Let $f : D \to \mathbb{R}$ be differentiable on some open interval $(a, b) \subset D$. Then, if f attains a maximum or minimum value at some point $c \in (a, b)$, then $f'(c) = 0$.

Proof. Let f attain a maximum at $c \in (a, b)$. Then $f(x) \leq f(c)$ for all $x \in (a, b)$. Since f is differentiable on (a, b) is follows that $f'(c)$ exists. Considering the right hand sided limit of the differential gives

$$f'(c) = \lim_{x \to c^+} \left(\frac{f(x) - f(c)}{x - c} \right) \leq 0.$$

Similarly, the left handed limit of the differential gives

$$f'(c) = \lim_{x \to c^-} \left(\frac{f(x) - f(c)}{x - c} \right) \geq 0.$$

Hence, $f'(c) = 0$. The proof for when c is a minimum is very similar and left as an exercise. □

This theorem was utilised by Fermat to solve so called optimisation problems. These are problems in which one is generally aiming to minimise or maximise a functions subject to some conditions. By realising that functions must have horizontal tangent lines at extremum values, Fermat utilised this to solve specific optimisation problems. Most likely you have exploited this theorem, or a small extension of this theorem, to great effect in previous calculus courses to find local minimums or maximums of specific functions. We will not go over more examples of these here, since this is not the purpose of introducing this theorem for us. Instead, we will utilise this theorem now to prove a new theorem that is in fact equivalent to the mean value theorem. This theorem is known as *Rolle's theorem* and it outlined below.

Theorem 4.2: Rolle's Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then, if $f(a) = f(b)$ there exists a point $c \in (a, b)$ such that $f'(c) = 0$.

Proof. By the extreme value theorem, it follows that f attains is bounds on $[a, b]$. If it attains its bounds at some point $c \in (a, b)$ then it follows by Theorem 4.1 that $f'(c) = 0$. Otherwise, if it attains its bounds at the end points it must be that f is the constant function on $[a, b]$. Hence, its derivative is zero everywhere on (a, b). □

Now whilst our primary purpose of introducing Rolle's theorem is to prove the mean value theorem, it can also be utilised to rove certain properties of cubic equations. Recall that the intermediate value theorem could be utilised to prove that

all cubic polynomials have at least one real root. Well, it turns out that Rolle's theorem can be utilised to show that specific cubic equations admit exactly one real root, and no more. This can often be useful in the area of *numerical analysis.* A specific example of this is given as follows.

Example 4.3: Roots of Cubics

Prove that the cubic polynomial $p : \mathbb{R} \to \mathbb{R}$ given by $p(x) = x^3 + x + 1$ has exactly one real root.

Proof. By the intermediate value theorem, it follows that p is at least one real root, say $a \in \mathbb{R}$. Suppose towards a contradiction that there exists another real root $b \in \mathbb{R}$ with $b \neq a$. Without the loss of generality, assume $a < b$. Now, since p is differentiable on $\mathbb{R}$, it follows that it is differentiable on (a, b) and continuous on $[a, b]$. Hence, by Theorem 4.2, there exists a $c \in (a, b)$ such that

$$p'(c) = 3c^2 + 1 = 0.$$

But no such c exists as $3c^2 + 1 > 0$ for all $c \in \mathbb{R}$. This is a contradiction and thus there exists exactly one real root. □

We are now finally in a position to prove the mean value theorem. The statement and proof of the theorem is given as follows.

Theorem 4.4: Mean Value Theorem

Let $f : [a, b] \to \mathbb{R}$. If f is continuous on $[a, b]$ and differentiable on (a, b), there exists a $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. The equation of the straight line going through the points $(a, f(a))$ and $(b, f(b))$ is given by

$$y = \left(\frac{f(b) - f(a)}{b - a}\right)(x - a) + f(a).$$

Now, the distance function $d : [a, b] \to \mathbb{R}$ between the curve $f(x)$ and the line y is given by

$$d(x) = f(x) - \left(\left(\frac{f(b) - f(a)}{b - a}\right)(x - a) + f(a)\right).$$

Clearly the distance between the curve and the line is zero at the end points. That is, $d(a) = d(b) = 0$. Additionally, by Theorem 3.3, and the algebraic properties of continuous functions, it follows that d is differentiable on (a, b) and continuous on $[a, b]$. By Theorem 4.2, there exists a $c \in (a, b)$ such that

$d'(c) = 0$. That is, there exists a $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

□

It is worth mentioning that prior to Cauchy, this theorem, much like the intermediate value theorem, was taking to be implicitly true. It was Cauchy who realised the importance of this theorem and the necessary steps required to prove it. One reason that this theorem was taken to be implicitly true is by thinking of functions as describing physical phenomena, such as particle trajectories. For example, suppose a function $f : [0, 1] \to \mathbb{R}$ describes the location of a particle at a specific time, where the unit of time is measure in hours. That is $f(0.5)$ is the location of the particle after half an hour. Then if the particle travels say 90 miles in one hour, then it must certainly have travelled at 90mph at some point along its journey. That is, its instantaneous speed must have equalled its average speed at some point along the journey. This is precisely what the mean value theorem is saying. This apparent obvious physical property of motion led people to take the mean value theorem as implicit true. A further comment to make is that the initial proof of the theorem assumed that f was differentiable on the closed interval $[a, b]$ as opposed to the open interval (a, b). This is also true of course, but the mean value theorem as stated above is slightly more general.
The mean value theorem is also frequently utilised to prove useful inequalities. These inequalities are often put to great effect in proving the convergence of specific sequences, series and even integrals, especially in courses involving statistics. Some examples demonstrating how the mean value theorem is used to prove inequalities is outlined in the follow example.

Example 4.5: Inequalities from Mean Value Theorem

(i) Let $f : [0, \infty) \to \mathbb{R}$ given by $f(x) = \cos(x)$. Then $\cos(x) \leq 1 + x$.

Proof. The function f is differentiable on $(0, x)$ and continuous on $[0, x]$ for each $x \in \mathbb{R}$. Hence, by Theorem 4.4 it follows that there exists a $c \in (0, x)$ such that

$$-\sin(c) = \frac{\cos(x) - 1}{x}.$$

Since $\sin(x) \leq 1$ for each $x \in \mathbb{R}$ it follows that

$$\frac{\cos(x) - 1}{x} \leq 1$$

$$\Rightarrow \cos(x) \leq x + 1.$$

Given that $x \in \mathbb{R}$ was arbitrary, we have that

$$\cos(x) \leq 1 + x$$

for each $x \in [0, \infty)$. □

(ii) Let $f : (0, \infty) \to \mathbb{R}$ given by $f(x) = x \ln(x)$. Prove that for each $k \in \mathbb{N}$ that

$$(k+1)\ln(k+1) - k\ln(k) \leq 1 + \ln(k+1).$$

Hence, deduce that

$$n\ln(n) \leq n + \ln(n!)$$

for each $n \in \mathbb{N} \setminus \{1\}$.

Proof. The function f is continuous on $[k, k+1]$ and differentiable on $(k, k+1)$ for each $k \in \mathbb{N}$. Hence, by Theorem 4.4, there exists a $c \in (k, k+1)$ such that

$$\ln(c) + 1 = (k+1)\ln(k+1) - k\ln(k).$$

Since f is an increasing function, it follows that

$$(k+1)\ln(k+1) - k\ln(k) \leq 1 + \ln(k+1).$$

Constructing the sum $\sum_{k=1}^{n-1}((k+1)\ln(k+1) - k\ln(k))$ gives

$$\begin{aligned}\sum_{k=1}^{n-1}((k+1)\ln(k+1) - k\ln(k)) &= n\ln(n) \\ &\leq n - 1 + \sum_{k=1}^{n-1}\ln(k+1) \\ &\leq n + \ln(n!).\end{aligned}$$

□

At this point, the discussion around the mean value theorem has been slightly disingenuous. The previous theorem that was referred to as the mean value theorem is not the one that Cauchy intially proved. Cauchy actually proved a slightly more general version that is sometimes now referred to as the *general mean value theorem*, or just *Cauhy's mean value theorem*. It turns out that it is this version we need to be able to relate differentiable functions to power series. The modern version of this theorem is outlined below.

Theorem 4.6: Cauchy's Mean Value Theorem

If $f : [a, b] \to \mathbb{R}$ and $g : [a, b] \to \mathbb{R}$ are both continuous on $[a, b]$ and differentiable on (a, b), then there exists a $c \in (a, b)$ with

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

Additionally, if $g'(x) \neq 0$ for each $x \in (a, b)$ and $g(a) \neq g(b)$, then the conclusion can otherwise be written as

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Proof. Define the function $h : [a, b] \to \mathbb{R}$ by

$$h(x) = (f(b) - f(a))g(x) - (g(b) - g(a))f(x).$$

By Theorem 3.3 and the algebraic properties of continuous functions, it follows that h is differentiable on (a, b) and continuous on $[a, b]$. Now, from Theorem 4.4, it must be that there exists a $c \in (a, b)$ such that

$$h'(c) = \frac{h(b) - h(a)}{b - a}.$$

Expanding both sides of this expression gives

$$\begin{aligned}(f(b) - f(a))g'(c) - (g(b) - g(a))f'(c) = \frac{1}{b-a}((f(b) - f(a))g(b) \\ - (g(b) - g(a))f(b) - (f(b) - f(a))g(a) \\ + (g(b) - g(a))f(a)) \\ = 0.\end{aligned}$$

Hence, there exists a $c \in (a, b)$ such that

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$$

□

This concludes the section on differentiation. The plan is to now exploit Cauchy's mean value theorem in order to derive a possible relation between differentiable functions and power series.

5 Power Series

For some time, the term power series has been used but not formally defined. We are now in a position to provide a definition of one. This is given as follows.

Definition 5.1: Power Series

A ***power series***, centered at 0, is a series of the form

$$\sum_{k=0}^{\infty} a_k x^k = a_0 + a_1 x + a_2 x^2 + \dots$$

where $a_k \in \mathbb{R}$ for each $k \in \mathbb{N}$.

The astute reader may have noticed this is not strictly a series as was defined in the series chapter. This was because a series required the terms in the partial sums to be real numbers. Here, the partial sums are all functions of x. This is not a drastic problem and can be repaired relatively easily by simply defining the notion of a sequence of functions. Unfortunately, due to time constraints, we are not able to explore the topic of function sequences. This instead will be discussed

in more detail in your third year complex analysis course. For now though, it is easiest to think of the series for specific values of $x \in \mathbb{R}$. In that case, it simply reduces to the case of a series as defined in chapter 3.
With this thought in mind, one then ponders the question, for what value of $x_0 \in \mathbb{R}$ does a power series converge? Well, it turns out that if a power series converges for some specific $x_0 \in \mathbb{R} \setminus \{0\}$, we can immediately deduce that it converges for a lot more values. This set of values for which the power series is convergent forms a beautiful symmetric region around zero with its explicit form outlined in the following theorem.

Theorem 5.2: Radius of Convergence

If a power series $\sum_{k=0}^{\infty} a_k x^k$ converges at some point $x_0 \in (0, \infty)$, then it converges absolutely for any $x \in (-x_0, x_0)$.

Proof. Suppose that $\sum_{k=0}^{\infty} a_k x_0^k$ converges with $x_0 \in (0, \infty)$. Then it follows by the divergence test that $\lim(a_k x_0^k) = 0$ and thus there exists an $M > 0$ such that $|a_k x_0^k| < M$ for all $k \in \mathbb{N}$. Now, for any $x \in \mathbb{R}$, it follows that

$$\begin{aligned} |a_k x^k| &= |a_k x_0^k| \left| \frac{x^k}{x_0^k} \right| \\ &< M \left| \frac{x}{x_0} \right|^k. \end{aligned}$$

The series given by

$$\sum_{k=1}^{\infty} M \left| \frac{x}{x_0} \right|^k$$

is geometric and thus converges if and only if $|x| < x_0$. By the comparison test, the series $\sum_{k=0}^{\infty} a_k x^k$ converges absolutely for any $x \in (-x_0, x_0)$. □

This theorem outlines that the sets on which a power series will converge must always take the form of either $(-R, R), [-R, R), (-R, R], [-R, R], \{0\}$ or $(-\infty, \infty)$ for some $R > 0$. In each case, apart from $[-R, R)$ and $(-R, R]$, the regions of convergence are symmetric about the origin. Given this, they can be thought of a *circles* in $\mathbb{R}$ centred about 0 and with radii R. It is due to this symmetry, that one refers to this value R as the *radius of convergence* of a power series. It's explicit definition is given as follows.

Definition 5.3: Radius of Convergence

For a power series $\sum_{k=0}^{\infty} a_k x^k$, the ***radius of convergence*** R is either

$$R := \sup \left(\left\{ |x_0| \in \mathbb{R} : \sum_{k=0}^{\infty} a_k x_0^k \text{ converges} \right\} \right),$$

or ∞ if the series converges for every $x_0 \in \mathbb{R}$.

Remark 5.4: Convergent Power Series

A power series $\sum_{k=0}^{\infty} a_k x^k$ with radius of convergence R can be written as a single real function $f : (-R, R) \to \mathbb{R}$ given by

$$f(x) = \sum_{k=0}^{\infty} a_k x^k.$$

If the series also converges at either R, $-R$ or both, then the domain on which this function is defined can be extended to either $[-R, R)$, $(-R, R]$ or $[-R, R]$.

Now, given a power series, it seems fairly clear that one would like to be able to calculate its radius of convergence, for this is the region on which the power series defines a function as according to Remark 5.4. Fortunately, we can simply exploit the series tests we developed in chapter 3. In fact, we only need one, namely the ratio test.

Theorem 5.5: Ratio Test Radius of Convergent

Let $\sum_{k=0}^{\infty} a_k x^k$ be a power series with $a_k \neq 0$ for all $k \in \mathbb{N} \cup \{0\}$. Then, if

$$\lim \left| \frac{a_{k+1}}{a_k} \right| = L$$

with $L \in (0, \infty)$, the radius of convergence is $1/L$. Otherwise, if $L = 0$, the radius of convergence is ∞.

Proof. Let $x_0 \in \mathbb{R}$ and

$$\lim \left| \frac{a_{k+1}}{a_k} \right| = L$$

with $L \in [0, \infty)$. Then by the ratio test, the series $\sum_{k=0}^{\infty} a_k x_0^k$ converges if

$$\lim \left| \frac{x_0^{k+1} a_{k+1}}{x_0^k a_k} \right| < 1.$$

By the algebra of limits it follows that

$$\lim \left| \frac{x_0^{k+1} a_{k+1}}{x_0^k a_k} \right| = |x_0| L.$$

Now if $L \neq 0$, the power series converges for values $|x_0| < 1/L$ and hence the radius of convergence if $1/L$. In the case when $L = 0$, it converges for every $x_0 \in \mathbb{R}$ and hence the radius of convergence is ∞. □

By utilising this theorem, we can now explicitly calculate the radius of convergence of a given power series. This is outlined as follows.

Example 5.6: Radius of Convergence

Consider the power series $\sum_{k=0}^{\infty} x^k/k!$. The radius of convergence of this power series is ∞.

Proof. Let $a_k = 1/k!$. Then,

$$\lim \left| \frac{k!}{(k+1)!} \right| = \lim \left(\frac{1}{k+1} \right) = 0.$$

Hence, by Theorem 5.5, it follows that the radius of convergence is ∞. This power series then defines a function $\exp : \mathbb{R} \to \mathbb{R}$ by

$$\exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

This is the definition of the ***exponential function***. □

6 The Success of Power Series

So we have now seen that we can define functions by restricting a given power series to the set of values for which it converges, which in turn is defined by the radius of convergence. We now consider the question of whether this function is differentiable and further, how its derivative is related to the initial power series. We might suspect that its derivative is simply given by differentiating term by term the power series. This comes from the assumption that the power series somewhat behaves like an ordinary summation. Contrary to many of the examples that demonstrated series do not behave like ordinary summations, it turns out that they do when acted on by derivatives. To prove this, it is first necessary to show that the power series in which each term has been differentiated once, has the same radius of convergence as the initial power series. This statement is outlined in the following theorem.

Theorem 6.1: Radius of Convergence of Differential

Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$ be a power series with radius of convergence R. Then the series $\sum_{k=1}^{\infty} k a_k x^{k-1}$ also has radius of convergence R.

Proof. Exercise □

Using this theorem, we can now relate the derivative of a function, written as a power series, to a new power series that has been differentiated term by term. Theorem 6.1 ensures that differentiating the power series term by term doesn't somehow give us a new series with a different radius of convergence. It it did, this would define a new function on a different domain. The theorem relating the derivative of a function to a power series is outlined below.

Theorem 6.2: Differentiating Power Series

Let $f : (-R, R) \to \mathbb{R}$ with

$$f(x) = \sum_{k=0}^{\infty} a_k x^k.$$

Then the function f is continuous on $(-R, R)$ and infinitely differentiable on $(-R, R)$. The derivatives of f are determined by

$$f'(x) = \sum_{k=1}^{\infty} a_k k x^{k-1}.$$

Proof. Omitted. □

Note that the proof of this theorem has been omitted, meaning it is not examinable. This is not due to it being overly complicated, but mainly due to the fact that we simply don't have the time to cover it. The most dedicated reader can have a go at proving this theorem, but it is considerably longer than other proofs given as exercises.

7 Taylor Series and Lagrange Theorem

We now arrive at the key theorems of this chapter, which are the culmination of all the work of the previous chapters. Before we outline these theorems, along with their proofs, we need one further definition on the topic of higher order derivatives. This is certainly something you have calculated before, and is a simply an iterative application of the definition of a derivative. It is given as follows.

Definition 7.1: Higher Order Derivatives and Smoothness

Let $f : D \to \mathbb{R}$. It is said that f is ***n-times differentiable on*** $I \subset D$ if $f, f', f'', \ldots, f^{(n)}$ exist at each point $c \in I$. The n-th derivative of f at the point $c \in I$, should it exist, is denoted by $f^{(n)}(c)$. A function that is infinitely differentiable on some interval $I \subset D$ is said to be ***smooth on*** I.

Finally we are in a position to outline Taylor's theorem. Here, we start with the assumption that our function is defined from a power series and given this, derive the values of the coefficients in the power series. This simply follows by exploiting Theorem 6.2.

Theorem 7.2: Taylor Series

Let $f : (-R, R) \to \mathbb{R}$ given by

$$f(x) = \sum_{k=0}^{\infty} a_k x^k.$$

Then the coefficients $a_k \in \mathbb{R}$ are given explicitly by

$$a_k = \frac{f^{(k)}(0)}{k!}$$

for each $k \in \mathbb{N} \cup \{0\}$.

Proof. By Theorem 6.2, it follows that

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)(n-2)\dots(n-k+1)a_k x^{n-k}.$$

Hence, $f^{(k)}(0) = k!a_k$. Dividing through by $k!$ gets the desired result. □

So given the assumption that a function can be written as a power series, we can explicitly find the coefficients of that power series. An example of this for the function $\sin(x)$ is demonstrated below.

Example 7.3: Taylor Series

Suppose that $\sin(x) = \sum_{k=0}^{\infty} a_k x^k$ with $x \in \mathbb{R}$. Then it follows by Theorem 7.2 that

$$a_k = \frac{f^{(k)}(0)}{k!}.$$

Then $f^{(2k)}(0) = 0$ and $f^{(2k+1)}(0) = (-1)^k$ for each $k \in \mathbb{N} \cup \{0\}$. Hence,

$$a_k = \begin{cases} \dfrac{(-1)^n}{(2n+1)!} & : \text{ if } k = 2n+1 \text{ for some } n \in \mathbb{Z} \\ 0 & : \text{ otherwise} \end{cases}.$$

Note the important point of Taylor's theorem is that we first assume that the function can be written as a power series. But, how do we even know this is a valid assumption in the first place? It is the purpose of Lagrange's theorem to answer this question and it does so by constructing the error between the function and its *Taylor polynomial of order* N. That is, the polynomial of order N with coefficients defined as in Taylor's theorem. Lagrange realised that by calculating the error between these two functions explicitly, and showing that this error goes to zero as N goes to infinity, this verifies that the function can be written as a power series. Lagrange managed to show that this error term always takes on a certain form and this is outlined explicitly in the following theorem.

Theorem 7.4: Lagrange's Remainder Theorem

Let $f : \mathbb{R} \to \mathbb{R}$ such that it is differentiable $N+1$ times at on $(-R, R)$, with $N \in \mathbb{N}$. Now, define $S_N : (-R, R) \to \mathbb{R}$ as

$$S_N(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_N x^N$$

with $a_k = f^{(k)}(0)/k!$ for each $k \in \{0, 1, \ldots, N\}$. This is called the ***Taylor polynomial of order*** N of f. Given $x \neq 0$ in $(-R, R)$, there exists a point c with $|c| < |x|$ where the error function $E_N(x) := f(x) - S_N(x)$ satisfies

$$E_N(x) = \frac{f^{(N+1)}(c)}{(N+1)!} x^{N+1}.$$

Proof. Let $x \in (0, R)$. Then the function $E_N(x)$ and x^{N+1} is continuous on $[0, x]$ and differentiable on $(0, x)$. Thus, by Theorem 4.6 there exists a $x_1 \in (0, x)$ such that

$$\frac{E_N(x)}{x^{N+1}} = \frac{E_N'(x_1)}{(N+1)x_1^N}$$

Now, the functions $E_N'(x)$ and $(N+1)x^N$ are continuous on $[0, x_1]$ and differentiable on $(0, x_1)$. Applying Theorem 4.6 again means that there exists an $x_2 \in (0, x_1)$ such that

$$\frac{E_N'(x)}{(N+1)x^{N+1}} = \frac{E_N''(x_2)}{(N+1)Nx_2^{N-1}}.$$

Continuing this $N+1$ times result in the existence of a $x_{N+1} \in (0, x_N)$ such that

$$\frac{E_N(x)}{x^{N+1}} = \frac{E_N^{N+1}(x_{N+1})}{(N+1)!}.$$

Setting $c = x_{N+1} < x$ gives the desired result since $E_N^{(N+1)}(x) = f^{(N+1)}(x)$. A similar proof can be constructed for when $x \in (-R, 0)$. This is left as an exercise. □

Note that Lagrange's remainder theorem does not claim that all infinitely differentiable functions can be written as Taylor series. It simply provides a value for the error term at each $N \in \mathbb{N}$ and x in the domain of the function. For us to be able to claim that a infinitely differentiable function can be written as a Taylor series, we must show that this error term $E_N(x)$ goes to zero as N goes to infinity for all x in the domain of the function. One such function we can prove this explicitly for is the sine function and is given as follows.

Example 7.5: Remainder Term for Sine

Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = \sin(x)$. It is the case that f is smooth on $\mathbb{R}$. Then for any $x \in \mathbb{R}$ it follows by Theorem 7.4 that there exists a c with $|c| < |x|$ such that

$$|E_N(x)| = \left| \frac{f^{(N+1)}(c)}{(N+1)!} x^{N+1} \right|.$$

Since the derivatives of f simply interchange between sine and cosine, and multiply by either 1 or -1, it follows that $|f^{(n)}(x)| \leq 1$ for every $x \in \mathbb{R}$ and

$n \in \mathbb{N}$. Thus,

$$|E_N(x)| \leq \frac{x^{N+1}}{(N+1)!}.$$

But we have shown in Chapter 2 that $\lim(x^n/n!) = 0$ for all fixed $x \in \mathbb{R}$. Hence,

$$\lim(|E_N(x)|) = 0$$

for each fixed $x \in \mathbb{R}$. It follows that

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}$$

for each $x \in \mathbb{R}$.

One may wonder then if all infinitely differentiable functions have a vanishing error term $E_N(x)$, as N goes to infinity for each x in the functions domain. Looking at the proof in Example 7.5, we saw that this was the case due to the fact $\sin(x)$, and all its derivatives, were bounded. So it would seem that this is certainly the case for functions of this type, and this is indeed true. However, despite people's belief for well over a century that all infinitely differentiable functions can be written as Taylor series, it was Cauchy who disproved this via the following counterexample.

Example 7.6: Smooth Does Not Imply Analytic

Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} e^{-x^{-2}} & : x \neq 0 \\ 0 & : x = 0 \end{cases}.$$

Now $f'(0)$ is given by

$$f'(0) = \lim_{x \to 0} \left(\frac{f(x) - f(0)}{x} \right) = \lim_{x \to 0} \left(\frac{e^{-x^{-2}}}{x} \right) = \lim_{x \to 0} \left(\frac{1/x}{e^{1/x^2}} \right).$$

This limit can be evaluated rigorously using L'Hopitals rule. For now, we will use the fact that exponentials grow much faster than polynomials and hence

$$\lim_{x \to 0} \left(\frac{1/x}{e^{1/x^2}} \right) = 0.$$

One spots that this would be the case for all the higher order derivative of f at 0. Namely, $f^{(n)}(0) = 0$ for all $n \in \mathbb{N}$. Hence, the function is differentiable at 0 infinitely many times, yet its *Taylor sum* is given by sums of zeroes. This demonstrated that not all infinitely differentiable functions can be written as power series, contrary to the belief that was held for over 100 years.

This counterexample demonstrates that the set of functions that can be written as power series, or Taylor series, is actually a strict subset of the set of infinitely differentiable functions. The collection of functions that can be written as power series are then given a special name, and are referred to as ***analytic functions***. You will explore these in much more detail in your complex analysis course where

weirdly, they are mush easier to classify.

This concludes the chapter on power series and differentiation as well as the entire course. As outlined in the opening remarks of this chapter, all the work of the previous chapters has culminated in the proof of Taylor's theorem and in turn Lagrange's remainder theorem. This gave us a way in which to prove specific smooth functions could be written as power series. You will find that this theorem, and multivariable extensions of it, are utilised to great effect in several of your future mathematics modules. Even today, 200 years after its formulation, it is the foundation of many areas of mathematics. Without it, such areas of approximation theory, data analysis, machine learning and numerical analysis would not exist. Further, specific differential equations would remain unsolved and thus make areas such as financial mathematics and mathematical physics highly restrictive in scope.